

Lab 2 - Agile Software Development

Coffee Maker Project

The School of Informatics at Leicester is purchasing a smart coffee maker. Unfortunately, the budget of the department was not sufficient to also purchase the software needed to make the coffee maker work. In this lab, your task is to adopt an agile approach to conduct a small software development project that will build the necessary software to provide the much-needed coffee to the department.

Project description

The coffee maker is very simple. It has internal containers for coffee and milk. It has two buttons, one to make an espresso and another one to make a latte. An espresso is just one unit of coffee without milk. A latte takes two units of coffee and one unit of milk. The department cannot afford to buy coffee for everyone, so they bought a coffee maker with a coin slot to receive payments for the coffees it makes. An espresso costs 1 pound and a latte costs 2 pounds. There are two additional buttons in the machine, one to display the current content of the containers (how much coffee and milk is left) and another one to do a cleanup of the machine (it should dispose of all the ingredients left in the container). To be able to dispense any beverage, the machine needs ingredients in its containers, therefore it also allows the office administrator to add the necessary ingredients (coffee and milk).

Task 1 - Identify user stories

First, read and understand well the CoffeeMaker project description, ask questions if you need clarification. Once it is all clear, create user stories for each of the expected features of the system you are going to implement.

The way in which user stories are phrased is subjective and may vary a lot from one person or team to another. Suppose we have the following five user stories:

- As a coffee drinker, I want to make an espresso.
- As a coffee drinker, I want to make a latte.
- As an office administrator, I want to clean up the machine.
- As an office administrator, I want to refill the machine.
- As an office administrator, I want to check how much the ingredients the machine has.

Task 2 - Estimate story points

After identifying the user stories, try to estimate the story points required using Planning Poker. You can create a point.poker session at <https://www.point.poker> to play the game.

Suppose your team have two members, your name is Tom, and your teammate name is Josh. The specific steps to play the Planning Poker game are as follows:

Prepare the game

- Click the above link and you will be greeted with following page.

Welcome to point.poker!

This is a simple online Planning Poker tool aimed for use by distributed collaborative teams.

Planning Poker (also called Scrum Poker, Pointing Poker and Point Poker) is a gamified technique for obtaining consensus-based estimations.

If you're here, you probably already know that. You can find out more about Planning Poker on [Wikipedia](#).

Why use point.poker?

- 100% free
- Fast & Easy-to-use
- No registration required
- Unlimited sessions, rounds and votes
- Responsive; create and join sessions on your desktop, tablet or smartphone

Create a  point.poker session

- Click the blue button to create a point.poker session and then enter your name.

Join Session

Enter your name below and get to voting!

Session ID: d87745c7-2d5f-4539-9c1e-db8d25e07b94

- Click blue button to join the game and you will see the following page.

Waiting for 1 more vote...

Tom

Next Round Show Votes

Vote

0	0.5	1	2	3	5	8	13
20	40	100	?	∞			

Invite

<https://www.point.poker/session/d87745c7-2d5f-4539-9c1e-db8d25e07b94>



- You can invite your teammate Josh by sending him the invitation link or asking him to scan the QR code. As below, Josh can join the game by opening the invitation link and entering his name:

Join Session

Enter your name below and get to voting!

Session ID: d87745c7-2d5f-4539-9c1e-db8d25e07b94

Play the game

- Now, your group can start your first round Planning Poker game.

Waiting for 2 more votes...

Vote

0	0.5	1	2	3	5	8	13
20	40	100	?	∞			

Invite

<https://www.point.poker/session/d87745c7->



- Given a specific user story, each group member needs to independently select proper story point from a list of Vote scores. After all group members have cast their votes, you can view the round summary by clicking **Show Votes**.

100%

All participants have cast their vote!

Vote

0	0.5	1	2	3	5	8	13
20	40	100	?	∞			

Invite

<https://www.point.poker/session/d87745c7->



Josh

✓

Tom

✓

100%

All participants have cast their vote!

Next Round

Show Votes

Josh	✓
Tom	✓

Vote

0	0.5	1	2	3	5	8	13
20	40	100	?	∞	🍌		

Invite

<https://www.point.poker/session/d87745c7->

Copy



Round Summary

Score Average

2

Suggested Score ?

2

Score	Count
3	1
1	1

Next Round

Hide Votes

Vote

0	0.5	1	2	3	5	8	13
20	40	100	?	∞	🍌		

Invite

<https://www.point.poker/session/d87745c7->

Copy



- The above round summary will show the suggested score for the given user story. The group members with high estimates and low estimates are given chances to offer their justification for their estimate and then the discussion continues. If a consensus is reached, the suggested score will be used as the story point for the given user story. Otherwise, you can start another round by clicking **Next Round** button.
- Repeat the games until a consensus is reached.

Task 3 - Build a GitHub repository

In this module, Git will be used in your group coursework. It is a distributed version control system that tracks changes in any set of computer files, usually used for coordinating work among programmers who are collaboratively developing source code during software development.

Please follow the provided Git tutorial to become familiar with some of the most used operations.

Now, create a GitHub repository to work on this project.

- If you do not have a GitHub account, create one at <https://github.com/join>. Each group will need a Github account!
- Next, one of you should create a new repository: <https://github.com/new> (only one repository for each group). Suggestion: consider using an appropriate `.gitignore` file.
- Open the command line and execute the following:

```
git clone https://<username>@github.com/<username>/CoffeeMaker.git
```

where `<username>` is your Github username and `CoffeeMaker` is the name of your repository. You should see an output like this:

```
1 $ git clone https://github.com/{your_username}/CoffeeMaker.git
2 Cloning into 'CoffeeMaker'...
3 remote: Enumerating objects: 3, done.
4 remote: Counting objects: 100% (3/3), done.
5 remote: Compressing objects: 100% (2/2), done.
6 remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
7 Unpacking objects: 100% (3/3), done.
```

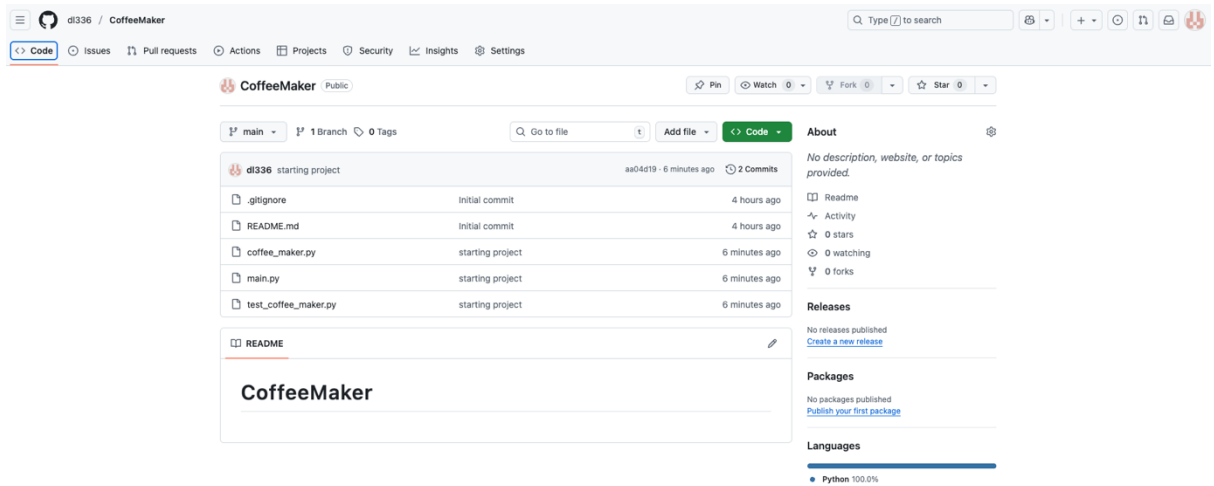
- Follow [this tutorial](#) to create a personal access token and then use this token as your password when pushing your changes to your GitHub repository.
- Add all the CoffeeMaker project code to the repository by moving the content of the CoffeeMaker directory into the newly cloned repository and executing the following:

```
git add .; git commit -m "starting project"; git push;
```

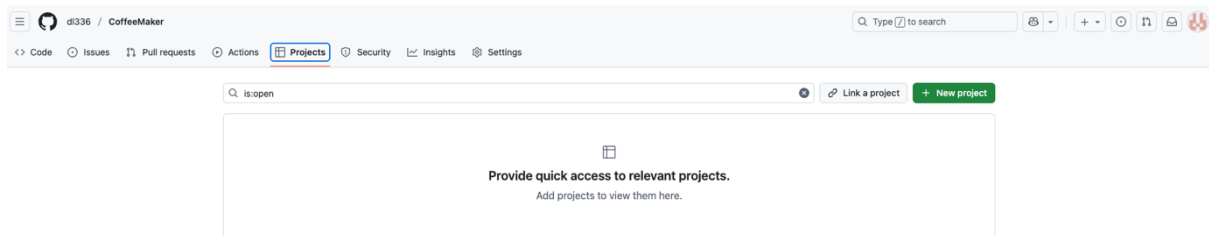
Task 4 - Create a scrum board

Create your scrum board within your Github repository as follows:

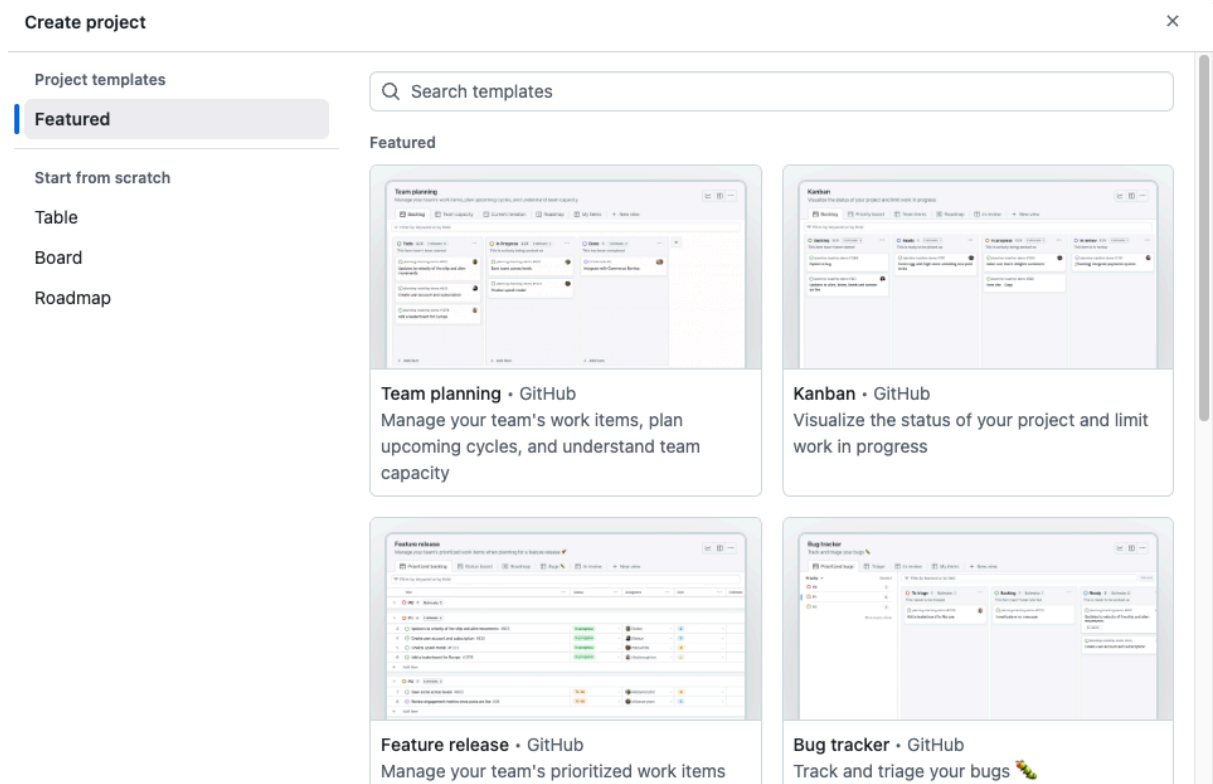
- Clicking `Projects` tab in your repo.



- Clicking **New project** button to create a new project.



- Choose the **Team planning** template.



- Change the project name to CoffeeMaker and then click **Create project**.

Team planning • GitHub

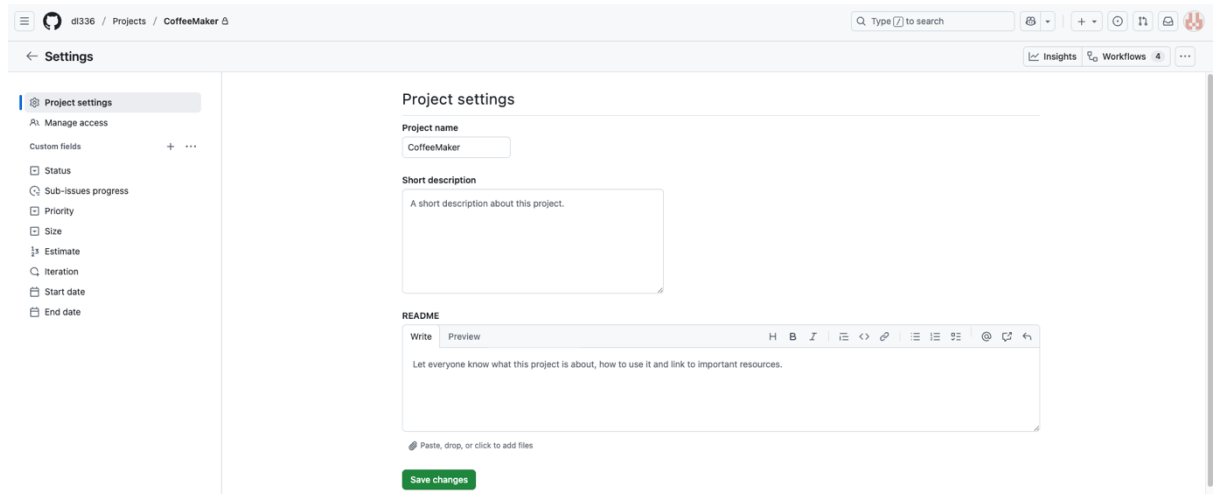
Manage your team's work items, plan upcoming cycles, and understand team capacity

Project name

CoffeeMaker

Create project

- Now, you need to change some settings to create the desired scrum board. Clicking the project name **CoffeeMaker** above Backlog and you will see:



- Choose **Manage access** tab from left panel. Here, you could invite **collaborators** (team members) by searching their usernames. You could also assign them different roles. The administrator in your project will be able to assign different user stories to different assignees.

Who has access

Private project

Only those with access to this project can view it.

Manage

Invite collaborators

Role: Write

Invite

Manage access

1 member

Type

Role

Find a collaborator

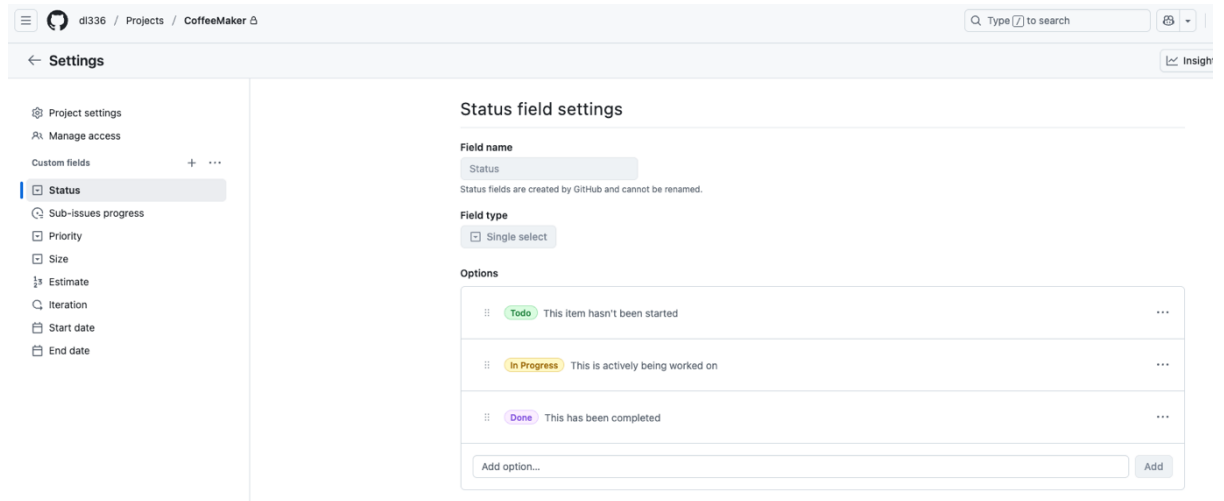
☐

dl336

Role: Admin

Remove

- Choose the **status** tab from left panel.



- Click ... next to Todo and choose Edit option.

Status field settings

Field name

Status

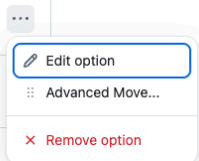
Status fields are created by GitHub and cannot be renamed.

Field type

Single select

Options

⋮	Todo	This item hasn't been started	⋮
⋮	In Progress	This is actively being worked on	⋮
⋮	Done	This has been completed	⋮
Add option...			Add



- Change Label text from Todo to Backlog and then Save.

- Next, click **Priority** tab from left panel.

- Under **Options** tab, click **...** next to P0 and choose **Edit option**. Change **Label text** from P0 to High and then hit **Save**. Similarly, change P1 and P2 to Medium and Low, respectively.

Priority field settings

[Delete field](#)

Field name

Field type

☒ Single select

Options

⋮	High	⋮
⋮	Medium	⋮
⋮	Low	⋮
Add option...		Add

- Next, click **Estimate** tab from the left panel and change the **Field name** from Estimate to Story point and hit Enter to save the change.

Story Point field settings

[Delete field](#)

Field name

✓ Saved!

Field type

☒ Number

- Next, click **Iteration** tab from the left panel and change the **Field name** from Iteration to Sprint and hit Enter to save the change.

Sprint field settings

[Delete field](#)

Field name

Field type

☒ Iteration

🕒 3 Active	📅 0 Completed	+ Add iteration	More options
Current	Iteration	🗑	
	2 weeks Sep 24 - Oct 07		
Planned	Iteration 2	🗑	
	2 weeks Oct 08 - Oct 21		
Planned	Iteration 3	🗑	
	2 weeks Oct 22 - Nov 04		
Save Reset			

- Click the current iteration name **Iteration** and change it to Sprint 1. Similarly, change **Iteration 2** to Sprint 2 and **Iteration 3** to Sprint 3. Finally, hit Save to save the changes. Here, you can also create or delete sprint or change its duration.

Sprint field settings

Delete field

Field name

Sprint

Field type

Iteration

3 Active

0 Completed

+ Add iteration

More options

Current	Iteration Sprint 1 2 weeks Sep 24 - Oct 07	
Planned	Iteration-2 Sprint 2 2 weeks Oct 08 - Oct 21	
Planned	Iteration-3 Sprint 3 2 weeks Oct 22 - Nov 04	

Save

Reset

Sprint field settings

Delete field

Field name

Sprint

Field type

Iteration

3 Active

0 Completed

+ Add iteration

More options

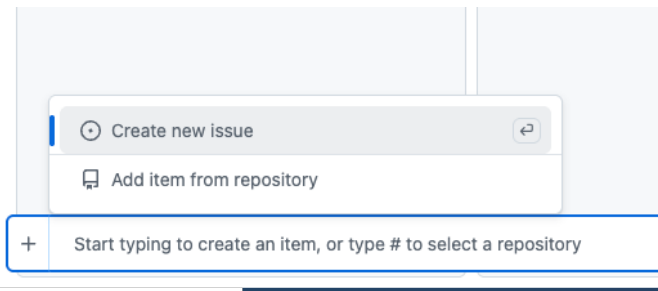
Current	Sprint 1 2 weeks Sep 24 - Oct 07	
Planned	Sprint 2 2 weeks Oct 08 - Oct 21	
Planned	Sprint 3 2 weeks Oct 22 - Nov 04	

Save

Reset

✓ Changes saved. Return to [CoffeeMaker](#).

- Click **+ Add item** on the bottom of the Backlog column and then click **+** button and choose **Create new issue** to input the first user story you identified in the previous step into **Add a title** box and then hit **Create**.



Create new issue in dl336/CoffeeMaker ×

Add a title *

As a coffee drinker, I want to make an espresso.

Add a description

Write Preview H B I

Type your description here...

Paste, drop, or click to add files

Assignee Label CoffeeMaker Milestone

☐ Create more

dl336 / Projects / CoffeeMaker

CoffeeMaker

Backlog Team capacity Current iteration Roadmap My items + New view

Filter by keyword or by field

Backlog 1 / 5 Story Point: 0

This item hasn't been started

CoffeeMaker #1

As a coffee drinker, I want to make an espresso.

In Progress 0 / 5 Story Point: 0

This is actively being worked on

Done 0 Story Point: 0

This has been completed

- Click the user story you just created within the Backlog column and enter the Assignees, Priority, Story point, and Sprint information. You could also click Create a branch under Development to build a specific branch for this user story.

As a coffee drinker, I want to make an espresso. #1

Edit

Open dl336/CoffeeMaker Public



dl336 opened 2 minutes ago

Owner

No description provided.

Create sub-issue

dl336 added this to CoffeeMaker 2 minutes ago

dl336 moved this to Backlog in CoffeeMaker 2 minutes ago



Add a comment

Write Preview

H B I

Use Markdown to format your comment

Paste, drop, or click to add files

Close issue

Comment

Assignees

No one - [Assign yourself](#)

Labels

No labels

Projects

CoffeeMaker

Status Backlog

Priority

High

Size

Choose an option

Story Point

3

Sprint

Sprint 1
Sep 24 - Oct 7
Current

Start date

No date

End date

No date

Milestone

No milestone

Relationships

None yet

Development

[Create a branch](#) for this issue or link a pull request.

Create a branch for this issue

Branch name

make-an-espresso

Repository destination

dl336/CoffeeMaker

Branch source

main

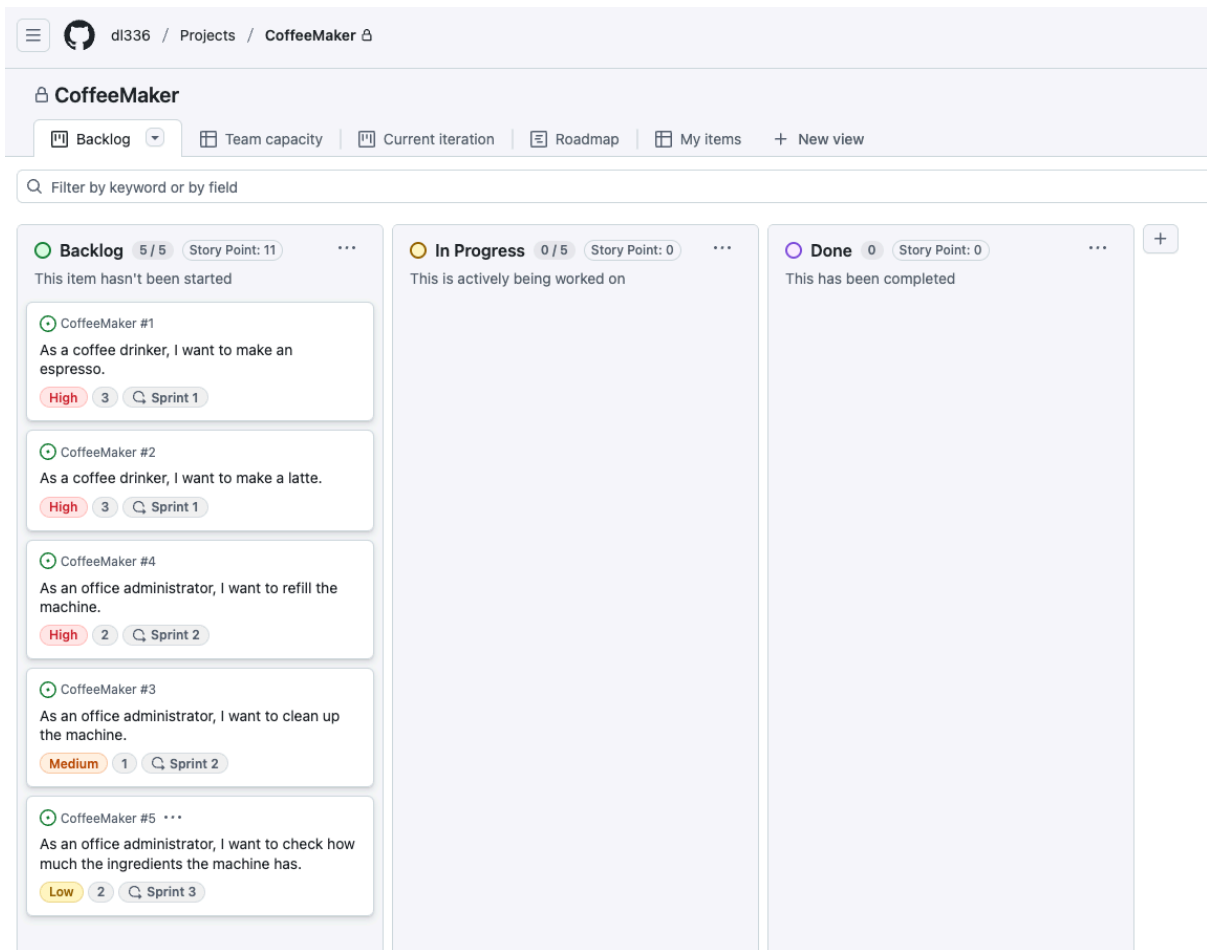
What's next?

☒ Checkout locally

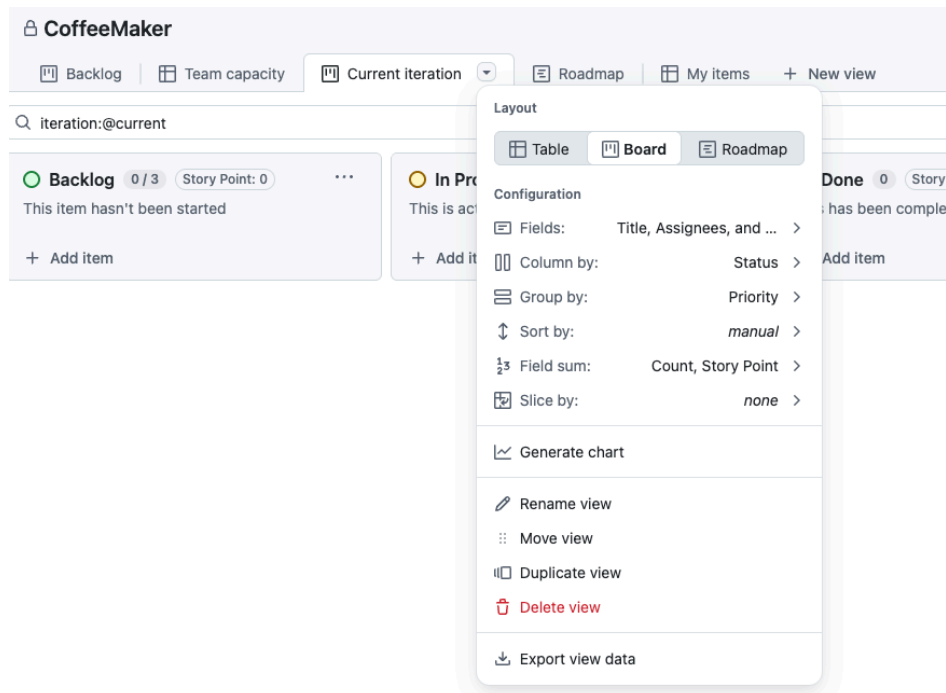
☐ Open branch with GitHub Desktop

Create branch

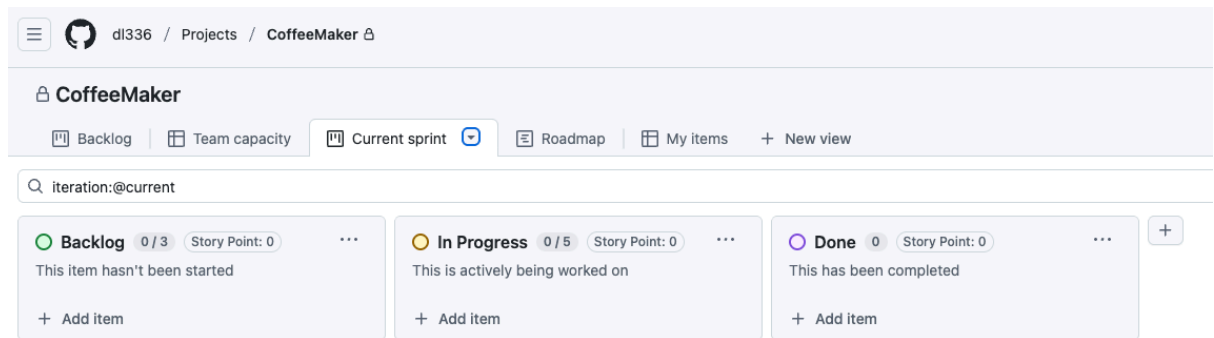
- Now, input all the user stories you identified in the previous step into the Backlog column. Below is an example, in which 3 sprints are created.



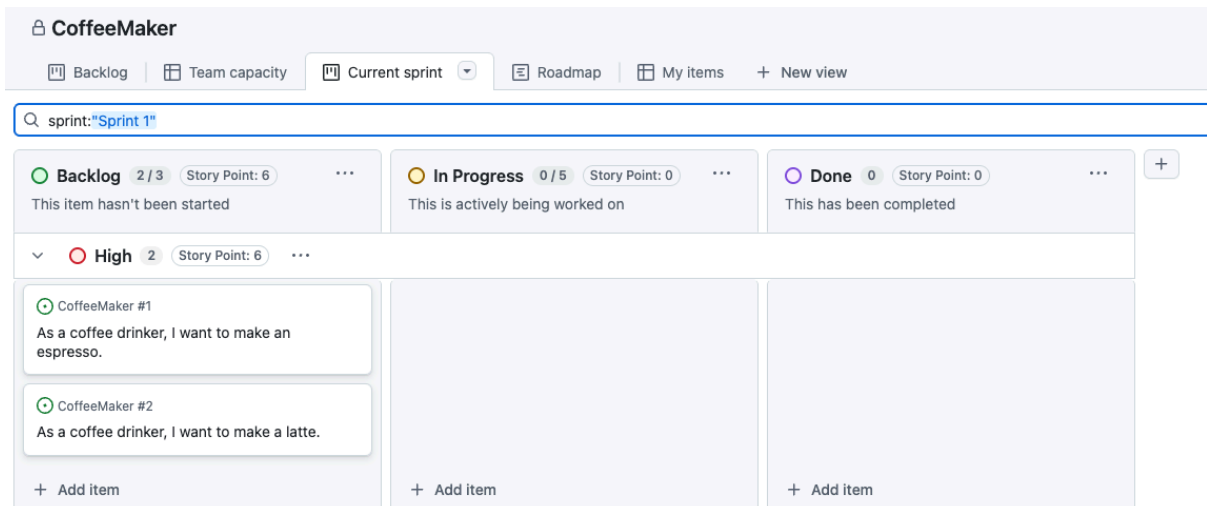
- Click **Current iteration** and then click the downwards triangle.



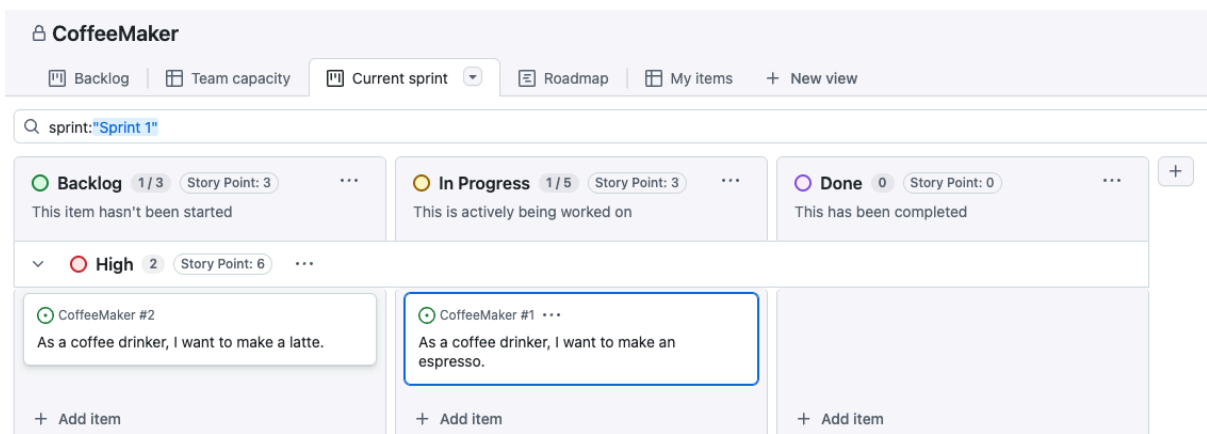
- Choose **Rename view** and change the **view name** from Current iteration to Current sprint and hit **Save** button.



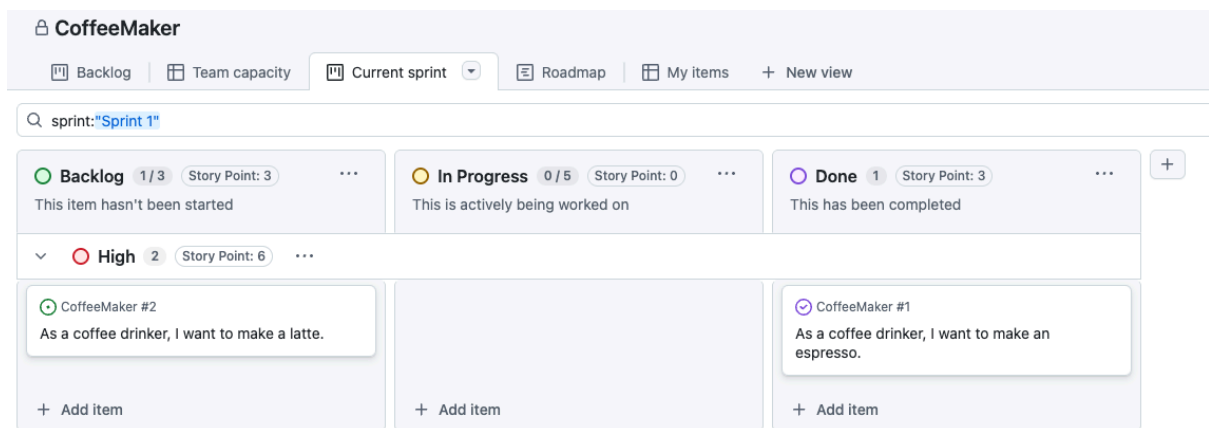
- Now, click the **Filter by keyword or by field** search bar and enter **sprint:"Sprint 1"** and then hit **Save** on the right side of the search bar, you will see all the user stories in Sprint 1. You can also check other sprints by changing field names, e.g., enter **sprint:"Sprint 2"** will show you all the user stories in Sprint 2.



- After you start to implement the first user story, you can drag it into **In Progress** column.



- If you finish this user story, you could further drag it into **Done** column.



Finally, for more information about managing projects in Github, you could refer to this [online tutorial](#).